D3 - Technical Methodology and Algorithm

Constrained Push-and-Settle Transport (CPST) for stockpile reconstruction and block modelling

1. Purpose and design philosophy

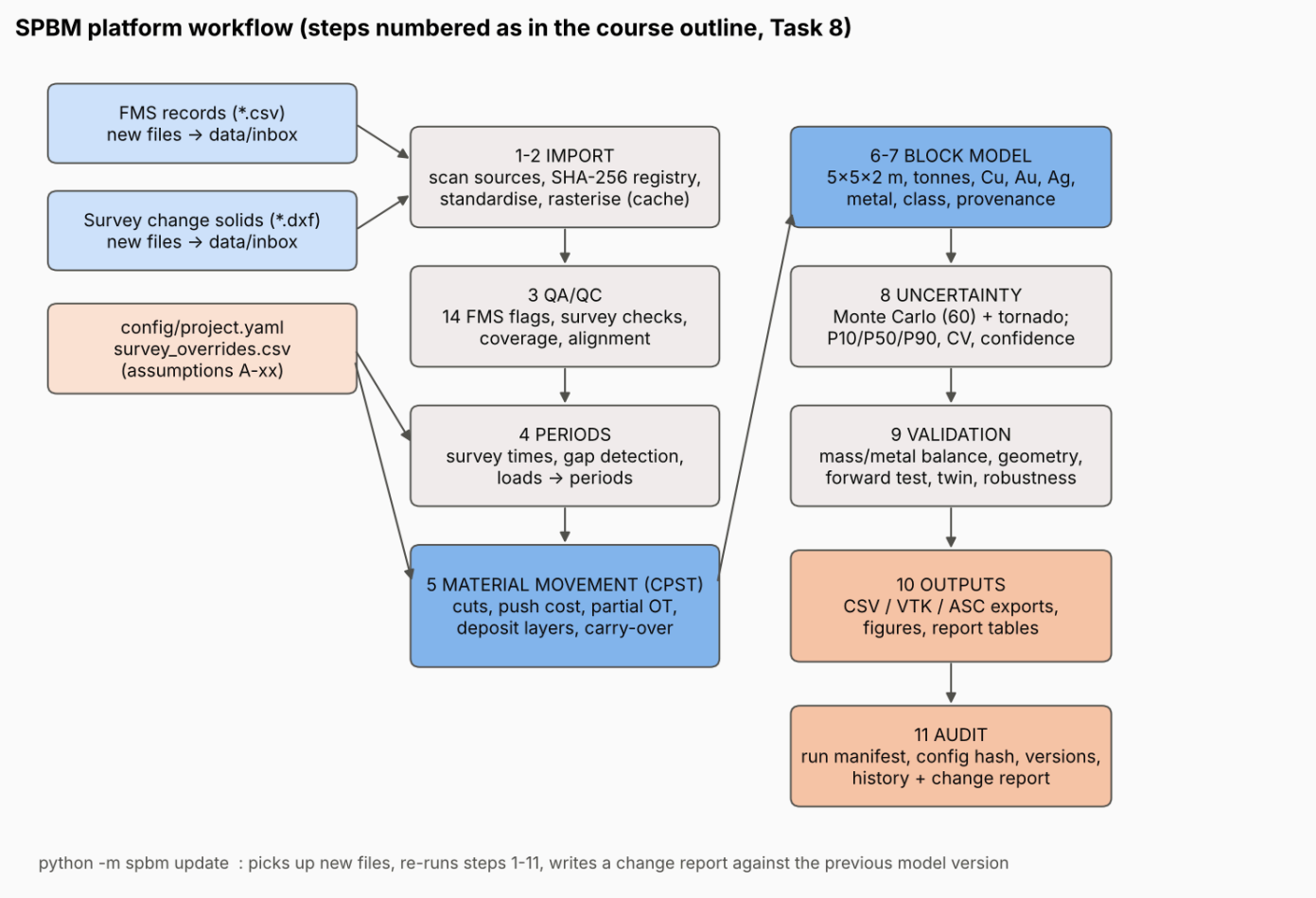
The movement of material from a truck dump point to its final place in the pile is never observed. Two kinds of observation constrain it:

- **where material came from**: FMS dump coordinates, tonnage and grade (*sources*);
- **where volume appeared**: the sequential survey change solids (*sinks*).

Physics-only rules (push to the crest, roll down at the angle of repose) can predict the shape of a deposit, but they predict *where* fill appears only moderately well. Our forward model places 40% of predicted volume inside the observed fill footprint on unseen periods (Section 9.3). A reconstruction used for inventory must therefore be **constrained by the surveys**. The design principle is:

> *Let the surveys decide **where** volume went, let the FMS decide **what** went there, and use a physically based transport cost (dozer push work over the real terrain) to decide **which** source filled **which** part of the observed fill, conserving mass and metal exactly.*

This is a classical earth-moving problem. Optimal transport was formulated by Monge (1781) to move *déblais* (cuts) to *remblais* (fills) with minimum work. We use its modern entropy-regularised form (Cuturi, 2013; Peyré and Cuturi, 2019), where the regularisation strength is a physically interpretable **mixing length**. We name the method **CPST: Constrained Push-and-Settle Transport**. A separate **forward physics model** (dozer push to crest + avalanche cellular automaton at the angle of repose) is used for prediction (temporal validation), for loads not yet surveyed (nowcast), and to generate known-truth data (synthetic twin validation).



Platform workflow

2. Data model

2.1 Grid and survey change solids

A regular grid of cell size Δ = 1 m (A-24) covers all data plus a 6 m margin. Its origin is snapped to a 10 m lattice so that grids of different resolution share block boundaries. Each survey file s is a closed triangulated solid (QA-S01). It is rasterised at cell centres c by barycentric interpolation. The highest face through a centre gives $\mathrm{top}_s(c)$ and the lowest gives $\mathrm{bot}_s(c)$. Cells crossed by no face are outside the solid. The implementation is vectorised (`spbm/grid.py::rasterize_triangles`) and independent of face winding.

2.2 State: stratigraphic columns

The stockpile state is a **column of layers** for every cell. Each layer ℓ has a bottom and top elevation and a vector of **extensive** properties:

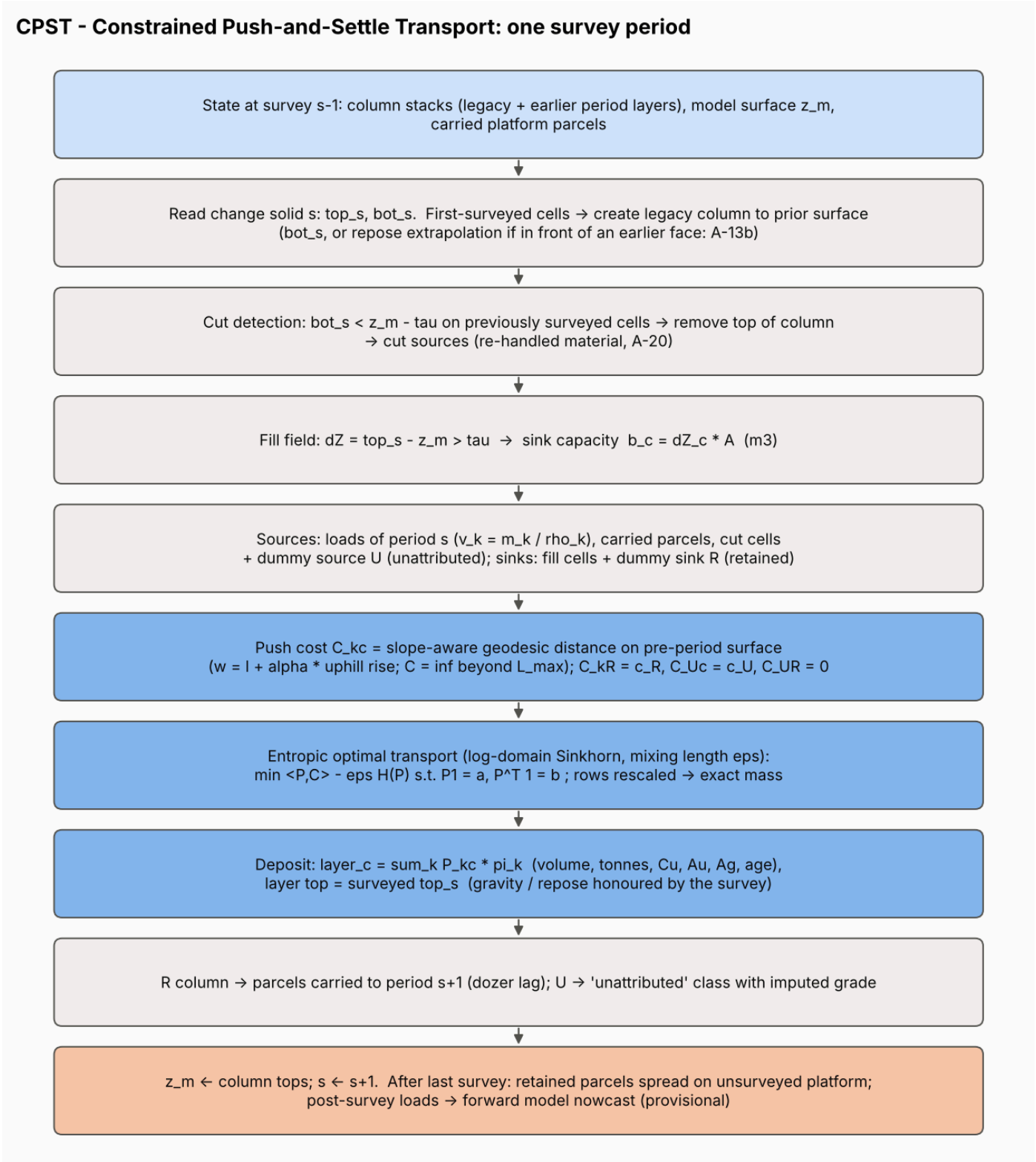
$$\boldsymbol{\pi}_\ell = (V,\ T_F, Cu_F, Au_F, Ag_F,\ T_U, Cu_U, Au_U, Ag_U,\ T_L, Cu_L, Au_L, Ag_L,\ \Sigma m g^2_{Cu},\ \Sigma m g^2_{Au},\ \Sigma m t,\ \Sigma m^2,\ T_R)$$

Here V is volume (m³) and T is tonnes. Cu is tonnes of copper ($mg_{Cu}/100$). Au and Ag are kilograms ($mg/1000$). The subscripts F, U, L and R mean FMS-tracked, unattributed, legacy and provisional. Every operation of the algorithm (deposit, split, cut, mix) is a **non-negative linear combination** of these vectors. Mass and metal are therefore conserved by construction, and grades are always derived as ratios (e.g. $g_{Cu} = 100\ \sum Cu / \sum T$), never averaged directly.

2.3 Periods

Survey s is assigned the time t_s = 23:59 on its (corrected) date (A-05). Period s is $(t_{s-1}, t_s]$. A load with time t_k belongs to period s if $t_{s-1} < t_k \le t_s$. Loads after the last survey are *provisional*. A period is a **gap period** if $t_s - t_{s-1} > 36$ h, or if more than 10% of previously surveyed cells inside solid s have bot_s above the model surface by more than τ (a missing intermediate survey).

3. The CPST algorithm (one period)



CPST algorithm

3.1 Prior surface and legacy material (first-surveyed cells)

For a cell surveyed for the first time, the prior elevation z_c^0 is the solid bottom $\mathrm{bot}_s(c)$, with one exception (A-13b). If the cell lies **in front of** a previously surveyed face, its prior elevation cannot exceed the

repose-angle continuation of that face:

$$z_c^{\inf} = \max(B_c,\ z_n - d_{cn}\tan\phi), \qquad z_c^0 = \begin{cases} z_c^{\inf} & \text{if } z_c^{\inf} < \mathrm{bot}_s(c) - \tau \\ \mathrm{bot}_s(c) & \text{otherwise} \end{cases}$$

Here n is the nearest surveyed cell, d_{cn} the distance, $\phi = 37°$ and B the fitted floor plane. "In front" is decided by the down-slope direction of the surveyed surface at n. The cell is in front if the vector $n \to c$ points down-slope ($|\cos| > 0.3$); otherwise the cell is behind (on the platform). The difference $\mathrm{bot}_s - z^0$ is fill placed during the record that no solid captured (toe-clipped solids, missing surveys). It joins this period's fill. A **legacy layer** from $\min(B_c, z_c^0)$ to z_c^0 is created with assumed properties (A-22).

3.2 Cuts (elevation decrease)

For previously surveyed cells, $\delta_c = \mathrm{bot}_s(c) - z_m(c) < -\tau$ indicates material removed since the last observation (A-20). The top $|\delta_c|$ of the column is removed. Its extensive properties $\mathbf{\Pi}_j$ (split proportionally within the cut layer) become a **cut source** j located at cell c.

3.3 Fill field (sinks)

$$\Delta Z_c = \mathrm{top}_s(c) - z_m(c), \qquad b_c = \Delta Z_c\,\Delta^2 \quad \text{for } \Delta Z_c > \tau$$

with the detection limit $\tau = k\sqrt{2}\sigma_z$ = 0.14 m (A-08, A-09).

3.4 Sources

- **Loads** k of the period: $v_k = m_k/(\kappa\rho_k)$ m³, located at the dump cell (for the 14 records with a default FMS position, the tonnage-weighted centroid of the period's other dumps, A-32). Properties per m³: $\boldsymbol{\pi}_k = \mathbf{\Pi}(m_k, g_k)/v_k$ (question A: *initial placement* is a point parcel at the reported X/Y; GPS uncertainty is handled by Monte Carlo, Section 7).
- **Carried parcels** from the previous period (A-23), at their original dump cell.
- **Cut sources** j (Section 3.2).
- **Dummy source U** (unattributed) with volume $V_F = \sum_c b_c$.

Sinks are the fill cells c, plus a **dummy sink R** (retained on the platform) with capacity $V_S = \sum_k v_k + \sum_j V_j$. The problem is balanced: $\sum a = V_S + V_F = \sum b$.

3.5 Push cost (questions B, C)

On the pre-period surface z (surveyed or inferred), a directed 8-neighbour graph is built. The edge weight from cell i to neighbour j is

$$w_{ij} = \ell_{ij} + \alpha \max(0,\ z_j - z_i)$$

where ℓ_{ij} = 1 or √2 m and α = 4 is the uphill penalty (A-16). The push cost is the geodesic (Dijkstra) distance $C_{kc} = \min_{\text{paths}} \sum w$, with $C_{kc} = \infty$ if $C_{kc} > L_{\max}$ = 90 m (A-17). Dummy costs: $C_{kR} = c_R$ = 90 m, $C_{Uc} = c_U$ = 90 m, $C_{UR} = 0$. Because $c_R + c_U > L_{\max}$, a load is always sent to reachable fill before being retained. U fills only the volume that available sources cannot (*partial optimal transport*).

Direction (B) is therefore the least-work route over the actual terrain: down-slope pushes are cheap, pushing over a crest and down the face is cheap, uphill is expensive. The survey decides which face sector actually received fill. *Distance* (C) is set by the capacity constraint: material stops where observed accommodation space exists, and the least-cost assignment is used. Movement beyond $L_{\max}$ is not allowed. Material that finds no reachable fill stays on the platform (R) and is carried to the next period (dozer lag).

3.6 Allocation: entropic optimal transport (questions E, F, G, H)

$$\mathbf{P}^{\star} = \arg\min_{\mathbf{P}\geq 0} \sum_{k,c} P_{kc}\,C_{kc} - \varepsilon H(\mathbf{P}) \quad \text{s.t. } \mathbf{P1} = \mathbf{a},\ \mathbf{P}^{\top}\mathbf{1} = \mathbf{b}, \qquad H(\mathbf{P}) = -\sum P_{kc}\,(\log P_{kc} - 1)$$

The solution is $P_{kc} = \exp\left((f_k + g_c - C_{kc})/\varepsilon\right)$. It is computed by **log-domain Sinkhorn** iterations (numerically stable with infinite costs):

$$f_k \leftarrow \varepsilon \log a_k - \varepsilon\,\mathrm{LSE}_c\left(\frac{g_c - C_{kc}}{\varepsilon}\right), \qquad g_c \leftarrow \varepsilon \log b_c - \varepsilon\,\mathrm{LSE}_k\left(\frac{f_k - C_{kc}}{\varepsilon}\right)$$

Iteration stops when $\sum_k |\sum_c P_{kc} - a_k| / \sum a < 10^{-6}$ (max. 5000 iterations). A final row rescaling $P_{kc} \leftarrow P_{kc}\,a_k / \sum_c P_{kc}$ makes **each source's volume exactly conserved**. The remaining column residual (≤ 8×10^{-7} relative in all periods) is reported as the numerical reconciliation error.

ε is the mixing length (A-18). As ε → 0 the plan becomes the work-minimising deterministic assignment (no mixing beyond what is geometrically forced). As ε → ∞ every source is spread uniformly over the period's fill, giving the "period-average" model. ε = 6 m represents GPS error, blade spill and rolling on the face. Its effect is quantified by the twin experiment, the tornado and the Monte Carlo.

3.7 Deposit (questions D, E, F, I)

Each fill cell receives one new layer from $z_m(c)$ up to $z_m(c) + \sum_k P_{kc}/\Delta^2$, which equals the surveyed $\mathrm{top}_s(c)$ to solver tolerance:

$$\boldsymbol{\pi}_{\mathrm{layer},c} = \sum_k P_{kc}\,\boldsymbol{\pi}_k + P_{Uc}\,\boldsymbol{\pi}_U, \qquad \Sigma m_c^2 = \sum_k (P_{kc}\,m_k/v_k)^2$$

Downslope behaviour (D) is honoured because the layer top is the surveyed surface: faces are at the measured angle of repose (median 36.7°), and the material that formed them, including what rolled to the toe, is assigned by the transport plan. Within one period the layer is vertically homogeneous (A-19). *Mixing* (E) is the volume-weighted combination of sources in the same cell. Grades follow from metal over tonnes, so mixing of loads with different densities is handled correctly. *Surface reconciliation* (F) holds because the column marginals equal the observed fill: the model surface equals the survey (Section 9.2). The unattributed U volume carries the imputed properties $\boldsymbol{\pi}_U$ (A-21).

3.8 Sequential updating (I) and closing the record

After deposit, z_m = column tops, the volume in column R becomes carried parcels, and the next period starts from this state. After the last survey:

1. Never-surveyed domain cells receive legacy columns up to the inferred pre-record surface (A-13).

2. Parcels still retained are spread (Gaussian kernel, σ = 6 m) over **unsurveyed** platform cells around their dump point.
3. Loads after the last survey are placed by the forward model (Section 4) and flagged *Provisional*. When the next survey is imported they are automatically reconciled like any other load.

3.9 Mass and metal conservation (G, H)

- Every source row of **P** sums exactly to the source volume, so no FMS tonne is created or lost.
- Cuts move extensive properties from columns to sources and back into columns.
- The dummy U adds volume only to the *unattributed* class; legacy volume is only in the *legacy* class. FMS totals are therefore checked separately from them.
- Verification: FMS tonnes, Cu, Au and Ag in the final block model differ from the FMS input by 3×10^{-12} (relative), i.e. floating-point round-off. This is enforced by the automated test `tests/test_core.py::test_end_to_end_mass_and_metal_conservation`.

3.10 Numerical stability (J)

Setting	Value	Effect tested
Grid cell	1 m (0.5 and 2.5 m tested)	Robustness table, Section 9.6
Detection limit τ	0.14 m (k = 1-4 tested)	Volume not moved: 96 m³ (0.2%)
Sinkhorn tolerance	10^{-6} (10^{-4}, 10^{-8} tested)	Column residual ≤ 8×10^{-7}
Log-domain Sinkhorn	always	No underflow with costs ≫ ε
Time step	one survey interval (data-driven)	Gap periods flagged
Block size	5 × 5 × 2 m (2 × 2 × 1 and 10 × 10 × 4 tested)	Tornado

Typical convergence takes 10-230 Sinkhorn iterations per period. A full reconstruction takes about 6 s.

3.11 Pseudocode

```
INPUT  surveys s = 0..S-1 (top_s, bot_s, t_s), loads k (x, y, t, m, g, rho), config
STATE  columns[c] = list of (z_bot, z_top, props);  z_m[c];  observed[c];  carried = []
B <- fit_floor_plane(toe cells)            ; domain <- hull(survey cells, dumps)
for s in 0..S-1:
    F <- cells with finite top_s
    for c in F not observed:               # first-surveyed cells
        z0 <- bot_s[c]
        if s>0 and front_of_earlier_face(c): z0 <- min(z0, repose_extrapolation(c))
        columns[c] <- [legacy layer B..z0]
    for c in F observed and bot_s[c] < z_m[c]-tau:
        cut_sources += remove_top(columns[c], z_m[c]-bot_s[c])
    fill <- {c in F : top_s[c]-z_m[c] > tau},  b_c <- (top_s-z_m)*A
    sources <- carried + loads(period s) + cut_sources
    a <- [v_k ..., V_fill],  b <- [b_c ..., V_sources]
    C <- geodesic_cost(surface, alpha, L_max) ; add dummy costs c_R, c_U, 0
    P <- sinkhorn_log(a, b, C, eps) ; P <- rescale_rows(P, a)
    for c in fill: append layer sum_k P[k,c]*pi_k + P[U,c]*pi_U to columns[c]
    carried <- {k : P[k,R] > 0}            # dozer lag
```

```
    observed[F] <- True
add legacy columns on never-surveyed domain cells (inferred surface)
spread carried parcels on unsurveyed cells; nowcast post-survey loads (forward model)
blocks <- integrate layers into 5x5x2 m blocks (exact partial-layer fractions)
```

4. Forward (predictive) physics model

Used where no survey constrains the result (temporal validation, nowcasting, synthetic truth). Code: `spbm/forward.py`.

1. **Crest detection.** On the surface smoothed with a 1.5 m Gaussian, a face cell has slope > 20° and at least 5 m of local relief within 11 m. Crest cells are gentle cells adjacent to a face and near the local maximum. Only crests within the surveyed area are used.
2. **Dozer push.** Each load is pushed to the nearest crest cell and released down-slope onto the face (release offset 0-4 m, calibrated), as a disc of radius 2-5 m (calibrated).
3. **Gravity / avalanche cellular automaton.** Only the newly placed (*mobile*) material may move. For each cell i and neighbour j, the excess is $e_{ij} = \max(0, z_i - z_j - \ell_{ij} \tan\phi)$. The cell releases $o_i = \min(\frac{1}{2}\max_j e_{ij}, \text{mobile}_i)$, distributed to neighbours in proportion to e_{ij}. This repeats until $\max o_i < 0.01$ m. Volume is conserved exactly (unit test). Slopes relax to ≤ φ.

5. Block model construction (Task 5)

- **Blocks**: 5 m × 5 m × 2 m, origin (511,290; 7,122,288; 4,474). Horizontal size: GPS error (σ ≈ 3 m) plus the mixing length (6 m) mean no information exists below about 5 m. Vertical size: 2 m gives 6-8 slices over the 13-15 m face, matching the vertical resolution of daily layers (median solid thickness 0.9-3.0 m). It also suits loader reclaim slices. Larger and smaller blocks were tested (tornado).
- **Discretisation / partial blocks**: layers are integrated exactly into blocks (the fraction of each layer's thickness inside each 2 m slice). No sub-blocking is needed: each block carries its **fill fraction** and true material volume; partial blocks are exported, not rounded.
- **Density**: per-load FMS density × κ for FMS material. Imputed density for unattributed fill. Mean FMS density for legacy. Block density = tonnes / volume (derived, never averaged).
- **Grade compositing / mixing**: tonnage-weighted via extensive metal. The within-block heterogeneity $s_{Cu} = \sqrt{\Sigma m g^2 / \Sigma m - \bar{g}^2}$ is reported.
- **Provenance**: a sparse load × block tonnage matrix built from the stored transport plans. It gives the effective number of loads $n_{\text{eff}} = (\sum t)^2 / \sum t^2$ and the top three contributing cycle IDs per block (traceability for reconciliation with mill results).
- **Uninformed regions**: legacy (pre-record) material and unattributed fill are kept in **separate tonnage fields**. A block's `info_class` is its dominant class. Geometry is flagged by `top_surveyed_frac`.

Fields exported per block: indices and centroid, dimensions, volume, fill fraction, tonnes, density, Cu %, Au g/t, Ag g/t, contained Cu (t), Au (kg), Ag (kg), tonnes by class, FMS-only grades, within-block SD, n_eff, top loads, mean placement date, surveyed-top fraction, Monte Carlo P10/P50/P90 and CV, information class and confidence class.

5.1 Confidence classification

Class	Rule
High	≥ 80% of tonnes FMS-tracked, Cu CV ≤ 10%, ≥ 50% of footprint has a surveyed top
Medium	≥ 50% FMS-tracked and Cu CV ≤ 20%
Low	≥ 20% FMS-tracked
Uninformed	< 20% FMS-tracked (legacy or unattributed dominated)

These are internal planning classes, **not** JORC/NI 43-101 resource categories. Public reporting would require a Competent/Qualified Person.

6. Information types

Quantity	Type
Dump X/Y, timestamp, payload, source grade, density	Measured (with stated errors)
Survey top/bottom surfaces in solid footprints	Measured
Survey time-of-day, file re-dating	Inferred from data
Floor plane, pre-record surface outside solids, front-of-face prior	Inferred
Legacy grade/density, unattributed grade, κ, φ in unsurveyed areas	Assumed
Location of each load inside the fill, block grades, uncertainty ranges	Model-generated

7. Uncertainty methodology (Task 7)

Monte Carlo (60 realisations, seed 413). Each realisation runs the full reconstruction with jointly sampled inputs:

Source	Sampling
Truck GPS error	X, Y + N(0, 3 m) per load
Tonnage measurement	× (1 + N(0, 0.08)) per load, × (1 + N(0, 0.03)) systematic
Assay / source grade	× (1 + N(0, 0.10)) per grade signature (Cu, Au, Ag independently)
Density / compaction	$\kappa \sim U(0.80, 1.05)$
Survey error	vertical bias N(0, 0.05 m) per survey
Survey timing	survey time ∈ {21:00, 22:00, 23:00, 23:59}
Degree of mixing	$\varepsilon \sim U(2, 15)$ m
Unknown dozer trajectories	$\alpha \sim U(1, 10)$
Movement distance	$L_{max} \sim U(50, 130)$ m

Source	Sampling
Missing data	unattributed grade drawn by tonnage-weighted bootstrap of the period's loads

Blocks are accumulated on the base block grid. For each block we report P10/P50/P90 and the coefficient of variation of Cu, Au and tonnes; for the whole stockpile we report P10/P50/P90 totals. **One-at-a-time sensitivity** (tornado) varies ε, α, $L_{\max}$, κ, σz, survey time, grid cell, block size and the A-13b switch individually.

8. Validation methodology (Task 6)

Area	Metric
Geometry	RMSE, bias, P1/P99, volume difference of model top vs latest surveyed top; fraction within τ; error map
Mass balance	FMS tonnes in vs block model (by class); unattributed and retained volumes per period
Metal balance	Cu, Au, Ag in vs block model
Temporal	Forward model calibrated on periods 1-5, tested on 6-9 (unseen). Metrics: Wasserstein-1 (earth mover's) distance between predicted and observed fill (m), capture ratio (share of predicted volume inside observed fill), centroid error. Baseline: no dozer push
Inversion skill	Synthetic twin: truth generated with the forward model (push jitter 6 m), observed with GPS noise (3 m), payload noise (5%), 15% missing FMS records and survey noise. Block Cu/Au RMSE and correlation vs truth for CPST variants and baselines
Robustness	Grid 0.5/1/2.5 m, Sinkhorn tolerance, detection limit: totals, top-bench grade, mean absolute block change

9. Implementation map

Function	Module
Source scanning, hashing, readers	`spbm/io.py`, `spbm/dataset.py`
Rasterisation	`spbm/grid.py`
Floor, domain, inference	`spbm/surfaces.py`
QA/QC	`spbm/qaqc.py`
Periods, alignment, growth	`spbm/growth.py`
CPST engine	`spbm/movement.py` (`reconstruct`, `sinkhorn`, `geodesic_cost`)
Forward model	`spbm/forward.py`
Block model, exports	`spbm/blockmodel.py`, `spbm/bmviz.py`
Validation	`spbm/validation.py`

Function	Module
Uncertainty / sensitivity	`spbm/uncertainty.py`
Audit trail	`spbm/audit.py`
Orchestration / CLI	`spbm/pipeline.py`, `spbm/__main__.py`

References

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